

Frederick (Tzujie) Huang

Curriculum Vitae

University of Texas Health Science Center at San Antonio
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RESEARCH INTERESTS

Modeling, analysis, and design of functional polymeric materials using machine learning (ML) for structure, property, and processing prediction. Broad topics of interest include machine learning for materials characterization, autonomous experimentation, informatics, and inverse design. Current work is in designing antimicrobial polymeric composites, validating ML predictions for dental composites, and mining literature for legacy materials data.

EDUCATION

North Carolina State University
B. S., magna cum laude, in Biomedical Engineering
Joint program with UNC School of Medicine

Raleigh, NC
2025

University of North Carolina at Chapel Hill
B. S., with Distinction, in Biomedical Engineering

Chapel Hill, NC
2025

RESEARCH EXPERIENCE

University of Texas Health Science Center at San Antonio
Comprehensive Dentistry Research Associate

San Antonio, TX
October 2025–present

- Validated preliminary machine learning (ML) model predictions for polymer composite formulations through ISO 4049 experimentation.
- Curated proprietary formulations as training data and engineered features for improving ML model performance. Built local LLM-assisted materials property data scraper backend for dental literature.
- Designed and built custom laboratory instruments to increase composite sample production, reducing discrepancies in degree of conversion.
- Developed an antifouling, remineralizing polymeric nanocomposite, coupling a quaternary ammonium silane functionalized matrix with phosphate mineral fillers for secondary caries prevention.
- Instructed operation and safety on lab tests, most commonly with Instron, Konica Minolta, and ThermoFisher instruments. Wrote data cleaning, visualization, and analysis pipelines for acquired mechanical testing data.

University of North Carolina at Chapel Hill
Biomedical Engineering Research Assistant

Chapel Hill, NC
August 2024–May 2025

- Contributed to study aimed at developing: 1) pentaomic extraction method and 2) a lipidomic integrity number (LIN), a sample quality metric analogous to established integrity measures (PIN, DIN, RIN/DV200) used in transcriptomic and genomic analysis, standardizing quality assurance in future bioinformatics studies.
- Subproject (*scholarship awarded*) analyzed LC-MALDI-TOF lipidomic datasets from *R. norvegicus* brain and *E. fuscus* liver specimens to characterize the differential effects of sample preparation conditions (drying and dissolution/mixing) on lipid degradation.
- Identified candidate lipid classes, primarily phosphatidylcholine (PC) and phosphatidylethanolamine (PE), exhibiting greatest sensitivity to sample preparation-induced chemical degradation, annotating data through Lipid-Search 5.
- Developed data visualizations to evaluate lipid abundance across preparation conditions, characterizing lipid classes with statistically significant fold-changes as candidates for LIN feature weighting.
- Contributed to rationale for LIN feature selection. Determined preparation-sensitive lipids appropriate in a computational sample quality metric, with goal of establishing SOPs for data acquisition.

University of North Carolina at Chapel Hill
Pharmacoengineering Research Assistant

Chapel Hill, NC
December 2023–August 2024

- Synthesized liposomal carriers and metal nanoparticles for theranostic applications. Developed procedural proficiency in liposome preparation and colloidal characterization with zeta potentials, SEM, and image analysis. Project ended at 8 months following departure of supervisor.
- Maintained EH&S compliance protocols for handling cytotoxic compounds and nanomaterials.

PROFESSIONAL EXPERIENCE

UT Health San Antonio

Technology Transfer Analyst

San Antonio, TX

June 2026–August 2026

- Developed early-stage IP patentability, state-of-the-art, and patent portfolio overview reports for pre-seed MedTech startup to guide design in gastroenterology hardware.

Pulpdent Corporation

Contracted Project

Watertown, MA

May 2026–August 2026

- Conducted testing for experimental polymer composites for dental restorations, analyzing batch production consistency for process refinement.
- Owned ISO 4049 compliant materials characterization and statistical analysis on experimental resin formulations to support commercial development.

SELECTED PROJECTS

Automated Literature Mining Pipeline | *Data Extraction, Plot Digitization*

March 2026–present

- Refactored local and cloud-based LLM backend for figure-caption extraction into an open-source scientific figure analysis framework, enabling automated reading of figure captions with textual/tabular context from born-digital PDFs.
- Utilized OpenCV for image segmentation of multipanel figures using whitespace projection-profile analysis with a YOLOv11 contour-detection fallback to separate complex scientific images.
- Ported PaddleOCR into workflow for figure classification to digitize graphical figures and capture time dependent features.
- Ran validation and quality-assurance tests to quantify panel-segmentation accuracy, caption coverage, and caption-to-figure correspondence with automated integrity checks on the exported feature tables, establishing a measurable baseline prior to downstream extraction.
- Adapted an open-source autonomous-materials-discovery codebase (Argonne Polybot, JACS 2025) to dental materials science domain.

Predicting Reusable Rocket Booster Landing Success | *sklearn, Data Mining*

January 2026–March 2026

- Scraped all launch data using SpaceX's Falcon 9 boosters since 2010 from SpaceX's REST API and Wikipedia to construct a traditional ML model predicting Stage 1 landing success.
- Scored prediction performance across varying regression and classification models, tuning parameters through GridSearchCV and yielding up to 88% accuracy evaluated via confusion matrices.

Antifouling Polymeric Composites | *Mechanical Testing and Analysis*

December 2025–August 2026

- Designed experimental composites to address secondary caries with quaternary ammonium silane (QAS) functionalized polymeric composites.
- Acquired and cleaned mechanical, optical, and thermal material property data for 80+ unique samples.
- Drafted all source code required for exploratory data analysis, visualization, and interpretation among 12+ experiments.
- Implemented statistical packages to elucidate key insights behind mechanical behavior affected by experimental nanocomposite formulations with QAS functionalized matrices.

Orthotic Brace for UNC Physical Medicine and Rehabilitation | *Rapid Prototyping*

August 2024–May 2025

- Conducted shadowing and clinical interviews to develop a chest brace for adolescent pectus carinatum patients, implementing on-the-fly calibration and real-time feedback following ISO 14971 and 13485 standards.
- Altered patient Haller index by an average of 0.2 with a projected 30% decrease in standard treatment time.

Simulating a *C. geographus* Inspired Insulin Analog | *PyMOL, ClusPro, Chimera*

August 2024–December 2024

- Investigated limitations in commercial insulin products and INSR/IGF1 receptor binding ability of cone snail venom proteins, engineering rapid pharmacokinetic properties into a thermally stable, single-chain insulin analog.
- Computationally predicted 12+ energy minimized candidate molecules and conducted *in silico* protein-protein docking, benchmarking designs against predicted structures from Google DeepMind's AlphaFold.
- Combed NIH's FASTA database to identify candidate bacteria-sourced homologs for potential scale-up production.

TEACHING EXPERIENCE

Course Name/Number	Enrollment	Type	Year
BMME209, Materials Science of Biomaterials	37	Ugrad	Spring 2025

PRESENTATIONS

- S. Mathur, K. S. Paye, **F. Huang**, M. N. Cole, M. J. Gilbert, M. K. Hederick. (2025). *The Brace Check—Data Collection and Monitoring for Pectus Carinatum Patients*, Lampe Joint Department of Biomedical Engineering Senior Design Symposium, Raleigh NC.
- **F. Huang**, C. Arciniega, B. Ehrmann, J. Macdonald. (2025). *Quality Management in Lipidomic Extractions to Develop a Lipidomic Integrity Number*, UNC-Chapel Hill Undergraduate Research Symposium, Chapel Hill NC.

SELECTED HONORS AND AWARDS

Abrams Scholar, <i>College of Engineering, North Carolina State University</i>	2025
IPEP Distinguished Scholar, <i>School of Medicine, University of North Carolina at Chapel Hill</i>	2025
Dean's List, <i>University of North Carolina at Chapel Hill</i>	2021–2025

SERVICE

Classical Violin Instructor, <i>Musical Empowerment at UNC-Chapel Hill</i>	2022–2025
Finance and Operations Officer, <i>UNC Mens Club Ultimate Frisbee</i>	2022–2025

SKILLS AND TECHNICAL COMPETENCIES

- **Tools:** Python, MATLAB
- **Packages:** Numpy, pandas, sklearn, Matplotlib, Seaborn, SciPy.
- **Software:** PyMOL, ImageJ, Onshape, Fusion360, KiCAD, Chimera, STARE, Office Suite.
- **Characterization:** Contact Angle, Indentation, Mechanical Load Frames, FTIR, SEM, Thermal Analysis.